

Closed Loop Stepper Motor Design With Encoder for Stall-Detection

Authors: Texas Instruments Incorporated (2020)

ABSTRACT

This application report describes the hardware and software design for closed-loop stepper motor control using incremental or absolute encoders. It details algorithm design for tracking error calculation, PI loop tuning, and real-time stall detection to prevent motor damage.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides an industry-standard benchmark for the malha-fechada implementation and AS5600 encoder integration on the EB15.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Texas Instruments Incorporated provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.